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## DOCUMENT 1 — ELECTROLYZER

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Component: PEM Electrolyzer

Purpose:

Convert electrical power into hydrogen for seasonal energy storage.

Inputs:

- Electrical power  $P_{el}$  [W]
- Water supply

Outputs:

- Hydrogen mass flow  $m_{H2\_dot}$  [kg/s]
- Waste heat flow  $Q_{heat}$  [W]
- Hydrogen outlet pressure  $p_{out\_Pa}$  [Pa]
- Outlet temperature  $T_{out\_K}$  [K]

Key Variables:

$\eta_{el}$  : electrical efficiency [-]

$p_{out\_Pa}$  : hydrogen outlet pressure [Pa]

$T_{out\_K}$  : electrolyzer outlet temperature [K]

Equations:

$$m_{H2\_dot} = \eta_{el} * P_{el} / LHV_{H2}$$

$$Q_{heat} = (1 - \eta_{el}) * P_{el}$$

$$p_{out\_Pa} = p_{out\_Pa}$$

$$T_{out\_K} = T_{out\_K}$$

Assumptions:

- Constant efficiency operation
- System-level energy balance used